

Semantic Multi-Channel Emotion Expression for Virtual Humans

Category: Research

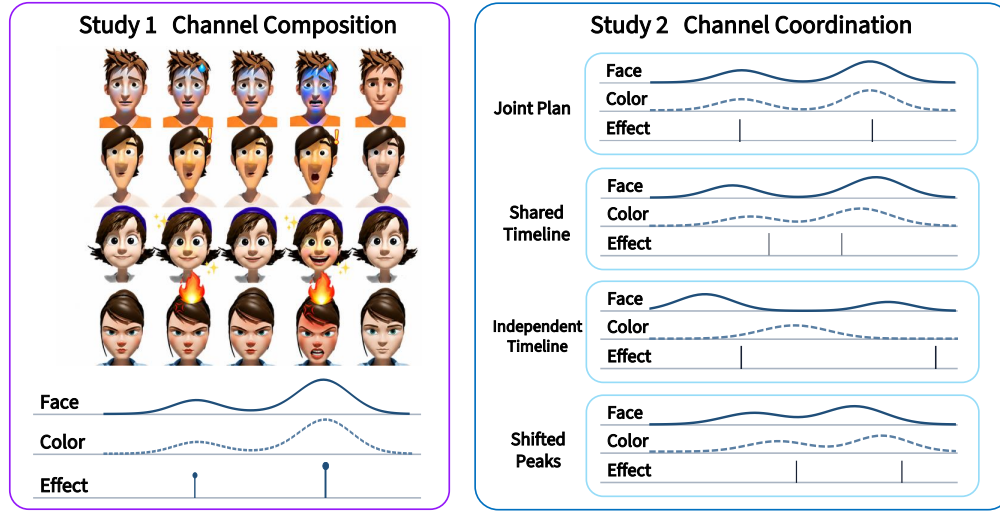


Figure 1: Overview of the two studies. Study 1 examines how facial expression, regional color, and symbolic effects jointly convey emotion. Study 2 compares four coordination strategies to investigate cross-channel coordination.

ABSTRACT

Speech-driven virtual humans support social interaction in VR, yet facial motion alone may not convey affect reliably. Although prior work has explored facial motion, facial coloration, and symbolic augmentation, these visual channels have largely been studied separately, leaving two related questions unresolved: **which channels should be combined**, and **how their content, intensity, and timing should be coordinated with speech**. To investigate these questions under controlled conditions, we develop an LLM-assisted, rule-constrained apparatus that maps speech semantics to structured control parameters for facial motion, color changes across six facial regions, and character-attached symbolic effects, while holding speech and lip synchronization constant. We conduct two VR user studies, each involving a distinct cohort of 36 participants. **Study 1** compares eight channel combinations across seven emotions and shows that the three-channel combination achieves the highest emotion recognition accuracy, whereas facial motion alone is rated most natural. **Study 2** compares four conditions crossing joint planning of channel content and intensity with coordinated timing of expressive peaks and shows that cross-channel coordination improves perceived naturalness while leaving recognition accuracy and confidence largely unchanged.

Index Terms: Virtual humans, social VR, emotional expression, facial animation, multimodal avatar control.

1 INTRODUCTION

Social virtual reality (VR) enables users in different locations to meet and communicate through virtual avatars. Since users see one another through these avatars, communication in social VR relies not only on speech but also on the nonverbal behaviors that avatars can display [33, 45]. However, speech content and lip-synchronized motion alone may not reliably communicate a speaker’s affect, particularly when facial tracking is incomplete or the avatar is stylized. Therefore, recent work has augmented speech-driven avatars with three broad classes of visual channels: emotional facial animation [8, 42, 39], affective facial coloration [10, 1, 4], and abstract

visual symbols [29, 18, 21, 46]. Although these approaches expand the available repertoire of affective cues, evidence from individual channels or task-specific combinations does not establish that their benefits remain additive when several channels are presented together. Co-occurring cues may become redundant or mutually inconsistent and compete for visual attention, potentially weakening affective legibility and perceived naturalness [6, 21]. Once multiple channels are available, the design problem shifts from cue accumulation to deciding which channels to combine and how to coordinate their content, intensity, and timing with speech. The effects of these composition and coordination decisions on emotion recognition and perceived naturalness remain insufficiently understood.

Realizing the joint potential of multiple visual channels in speech-driven emotional expression requires separating two related design questions. **(1) Which visual channels should be combined?** Facial motion, regional facial color, and symbolic effects each communicate affect through distinct visual mechanisms, yet their effectiveness in isolation does not establish whether they remain complementary when presented over the same utterance. The central issue is therefore not simply how many cues to add, but which combinations improve emotion legibility without compromising perceived naturalness. **(2) How should the selected channels be coordinated?** Even a well-chosen combination may appear incoherent if its channels encode inconsistent affective content, use poorly calibrated intensities, or unfold along disconnected timelines. Effective multi-channel expression therefore also requires coordinating what each channel conveys, how strongly it is expressed, and when it unfolds relative to speech.

To investigate these questions, we implement an LLM-assisted, rule-constrained apparatus that holds the input speech and neutral lip motion constant across conditions while independently manipulating three visual channels. The speech waveform is processed in parallel by an audio-conditioned LLM-based planner and a Speech2Lip module. The planner analyzes the speech and, in controlled mode, receives the target emotion as metadata separate from the spoken content. It generates a joint stage-level plan with shared temporal boundaries and channel-specific controls for emo-

tional facial motion, regional facial color, and avatar-attached effects. Deterministic rules validate the plan and bound its parameters before execution. Speech2Lip module uses NeuroSync [2] to generate frame-level ARKit blendshape coefficients [3], retaining only those associated with lip and jaw articulation to form the neutral lip-motion sequence. FaceMotionFusion module preserves this sequence for phoneme-critical controls, blends it with planned emotional motion for expressive mouth controls using speech-energy-dependent weights, and assigns the remaining facial controls from the emotional plan. The color plan parameterizes six facial regions, whereas the effect plan schedules symbolic graphics and particles with explicit intensity and timing. The resulting facial, color, and effect control streams are delivered to a physical renderer Unreal Engine 5.8 (UE5.8) [14] through LiveLink [12] and Open Sound Control (OSC) [13]. For Study 1, a binary mask independently gates the three planned channels while speech and neutral lip motion remain identical across conditions. The primary configuration uses a joint plan with shared stage boundaries. Channel-local planning, independent stage segmentation, and altered peak timing appear only in Study 2 comparison conditions.

Using this apparatus, we conduct two within-subject VR experiments, each involving a distinct cohort of 36 participants. **Study 1 (The Channel Composition Study)** uses a full-factorial design in which facial motion, regional facial color, and symbolic effects are independently enabled or disabled, yielding all $2^3 = 8$ channel combinations. Across seven emotions, participants identify the expressed emotion, report their judgment confidence, and rate the perceived naturalness of each presentation under otherwise matched speech, lip motion, avatar, and viewing conditions. The three-channel combination achieves the highest recognition accuracy, whereas facial motion alone receives the highest naturalness ratings, revealing a trade-off between affective legibility and perceived naturalness. **Study 2 (The Channel Coordination Study)** keeps all three channels active and compares four matched planning and timing conditions designed to isolate the effects of shared semantic stages, joint content-and-intensity planning, and temporal alignment of expressive peaks. It retains the same seven emotions and dependent measures, thereby separating the effects of coordination from changes in channel availability. Channel coordination improves perceived naturalness while leaving recognition accuracy and judgment confidence largely unchanged. Together, the studies distinguish two perceptual roles: channel composition shapes the trade-off between affective legibility and perceived naturalness, whereas coordination improves naturalness without materially altering emotion legibility.

Our contributions are summarized as follows:

- We develop an LLM-assisted, rule-constrained apparatus for independently controlling facial motion, regional color, and symbolic effects over fixed speech and neutral lip motion.
- We characterize all combinations of the three channels under matched VR conditions, finding that the three-channel combination maximizes recognition accuracy whereas facial motion alone is rated most natural.
- We isolate the effects of coordinating content, intensity, and timing across channels, finding improved naturalness while recognition and confidence remain largely unchanged.

2 RELATED WORK

2.1 Visual Affective Channels for Virtual Humans

Research on visual affective channels in social VR has expanded from anatomical facial motion to appearance-based facial color and symbolic effects. Each line has broadened the expressive range of virtual humans while introducing distinct perceptual constraints.

2.1.1 Facial Motion

Early speech-driven facial-animation research focused on mapping speech acoustics to plausible 3D facial motion, with subsequent advances improving temporal context modeling and motion priors [22, 15, 56]. Although affective variation appeared in early models [22], later emotion-aware systems made it a primary and explicitly controllable factor. EMOTE [8], EmoTalk [42], and Emotional Voice Puppetry [39] separated articulation from affective variation to different degrees and introduced controls over emotion category, intensity, identity, or speaking style. Although these methods make affective facial motion increasingly controllable, controllability alone does not ensure perceptual effectiveness in VR. In immersive viewing, facial motion is not a binary cue: its effects depend on which regions move and how strongly. Visible eye and mouth motion can increase social presence [24], whereas partial face animation can alter embodiment and uncanniness [28]. Region-specific exaggeration shows a similar dependency: strengthening weak upper-face cues can improve perceived realism for some emotions, while excessive mouth motion can reduce it [53]. Facial motion thus provides an anatomical foundation for speech-driven expression, but its perceptual contribution is shaped by motion coverage and the regional allocation of intensity.

2.1.2 Facial Color

Whereas facial motion conveys affect through deformation, facial color provides an appearance-based signal without requiring additional geometry. Early graphics research established how such cues could be rendered, from physiological manifestations such as blushing to dynamic skin-appearance models parameterized by hemoglobin and melanin [10, 20]. The central perceptual question is whether color merely reinforces facial deformation or conveys partly independent affective information. Perceptual studies show that facial color interacts with expression shape in emotion judgments [37], while spatially diagnostic color patterns can communicate emotion even without corresponding muscle motion [4]. Facial color can therefore contribute information that is not fully available from facial motion alone. Yet this independence does not mean that stronger or more dynamic color changes are uniformly more effective. Within natural skin-color ranges, hemoglobin and melanin modulations affect different emotions differently [23]. Dynamic transitions likewise do not consistently outperform static presentations, as judgments can instead be dominated by the terminal color [44]. Facial color thus forms a complementary appearance channel, but its perceptual contribution depends on the target emotion, the spatial distribution and magnitude of the color change, and the state at which the transition ends.

2.1.3 Symbolic Effects

Symbolic effects externalize reactions or otherwise hidden states through conventional graphics that can remain legible when facial detail or tracking is limited [45]. Within this design space, avatar-attached biosignal visualizations expose otherwise hidden physiological states [29], whereas emoji-like reactions and detachable expressions make communicative intent more explicit [18, 46]. This explicitness can facilitate identification, but the same visual salience can reduce naturalness, distract from the avatar, or add redundant information when combined with embodied cues [29, 18]. Moreover, biosignal displays, deliberate reactions, and automatically selected affective effects encode different semantic targets, so a visually legible symbol is not necessarily coherent with the accompanying face or speech. Symbolic effects therefore extend expression beyond the facial rig, but their usefulness depends on vocabulary, placement, intensity, and timing. Together with the conditional effects of facial motion and color, these limitations motivate controlled study of which channels should be combined and how their outputs should be coordinated.

2.2 Visual Affective Channels Composition

Composing affective channels is not simply a matter of accumulating cues. Face–voice studies [35] show that observers integrate concurrent signals and weight the clearer source more heavily when the other is ambiguous. Neural and divided-attention studies [9, 51] suggest that this integration emerges early and persists under limited attention, while incongruent body context can bias facial judgments [36]. Although this evidence spans sensory modalities, it identifies four outcomes relevant to concurrent visual channels: complementarity, redundancy, conflict, and dominance. These outcomes depend on cue clarity, congruence, and salience rather than cue count alone. Embodied-agent and immersive studies [7, 48] reveal similar trade-offs. Congruent face–posture or voice–body–context configurations can improve emotion recognition and agent evaluations, whereas neutral or conflicting channels can impair informative signals and trust-related judgments [31, 6]. Appearance-based and symbolic augmentations are similarly conditional: autonomic cues benefit selected emotions, biosignals trade arousal legibility for distraction, and emoji-like reactions may increase explicitness at the cost of naturalness or provide no advantage when combined with gestures [11, 29, 18]. Recent VR systems [21, 41] extend this design space through audiovisual and bioresponsive effects, but remain based on task-specific channel configurations.

Prior work shows that affective channels may complement, duplicate, or conflict with one another, yet heterogeneous agents, tasks, and channel sets obscure the effect of composition. To isolate this effect, we conduct **Study 1 (The Channel Composition Study)**: a controlled full-factorial comparison that independently activates facial motion, facial color, and symbolic effects while holding speech and neutral lip motion constant, measuring emotion recognition, judgment confidence, and perceived naturalness.

2.3 Cross-Channel Planning and Coordination

Once expressive channels are selected, coordination determines each channel’s content, intensity, and timing relative to speech and the others. Early architectures [5, 26] framed this as a representation-and-realization problem, mapping linguistic analyses to synchronized verbal and nonverbal behaviors. SAIBA [25] separated intent planning, behavior planning, and realization, while BML [50] specified cross-behavior synchronization points. Emotion-oriented models [40] extended this approach to temporally structured multimodal expression, while modular realizers [47, 49, 43] enabled real-time execution. These systems established how coordination could be specified and realized, but not how alternative policies affect perception. Coordination, however, entails more than simultaneity. Gesture studies [30, 32, 54] show that movement landmarks align with distinct prosodic units and that sensitivity to temporal offsets depends on shift direction and prosodic emphasis. Reciprocal facial timing likewise affects perceived agent responsiveness [27]. Speech-driven systems therefore combine semantic and prosodic analyses to schedule facial expressions, gaze, head motion, and gestures [34], reflecting the semantic, pragmatic, and temporal relations between speech and gesture [52]. Recent LLM-based methods [19, 16, 17] support context-sensitive gesture selection from utterance semantics, but remain gesture-centered and typically implement a single integrated policy.

Despite these advances, existing systems typically determine content, intensity, temporal structure, and alignment within a single pipeline, leaving the perceptual contribution of each coordination mechanism unclear. To isolate these effects, we conduct **Study 2 (The Channel Coordination Study)**, which compares four conditions crossing joint planning of channel content and intensity with coordinated timing of expressive peaks.

3 METHOD

3.1 Overview

Given a speech waveform a , our pipeline represents avatar expression through three complementary visual channels: facial motion, regional facial color, and symbolic effects. As illustrated in Fig. 2, the speech input follows two parallel branches: the Speech2Lip module produces the neutral lip-motion sequence $\mathbf{L}_{1:N}$ as an articulation baseline, while the LLM-based planner, instantiated with GPT-5.6 Sol [38], uses the utterance semantics and timing to coordinate the content, intensity, and temporal organization of the three visual channels in a shared stage-level plan S . Because the planned emotional facial motion may overlap with articulatory mouth motion, the facial-motion fusion module integrates it with $\mathbf{L}_{1:N}$ while preserving speech-critical motion, yielding the complete facial-control sequence $\mathbf{F}_{1:N}^{\text{full}}$. Finally, $\mathbf{F}_{1:N}^{\text{full}}$, the regional facial-color sequence $\mathbf{C}_{1:N}$, and the symbolic-effect schedule $\mathcal{E}$ are synchronized and transmitted to Unreal Engine 5.8 (UE5.8) through Live Link [12] and OSC [55, 13], where they are jointly rendered as the final multi-channel avatar animation.

3.2 LLM-based Multi-Channel Planner

Neutral lip motion provides the articulatory basis of speech, but it does not determine how affect should be expressed across concurrent visual channels. We therefore use an LLM-based planner, implemented with GPT-5.6 Sol [38], to interpret the utterance-level semantic and temporal context and generate a coordinated plan for facial motion, regional facial color, and symbolic effects. In open-ended use, the intended affect is inferred from the utterance; in controlled use, the planner can instead be conditioned on an externally specified target emotion that remains separate from the audible speech. In both cases, the planner operates at the level of semantically meaningful stages rather than directly producing frame-level animation.

We represent the resulting plan as an ordered sequence of stages,

$$\Gamma = \{\gamma_k\}_{k=1}^K, \quad \gamma_k = ([t_k^s, t_k^e], \eta_k, \iota_k, \phi_k, \chi_k, \varepsilon_k), \quad (1)$$

where $[t_k^s, t_k^e]$ denotes the temporal extent of stage k , η_k denotes its affective state, and ι_k denotes its expressive intensity. The channel-specific variables $\mathbf{f}_k$, $\mathbf{c}_k$, and $\mathcal{E}_k$ describe the planned facial motion, regional facial color, and symbolic effects, respectively. This representation captures the emotional progression of an utterance while keeping the three channel outputs explicitly separable.

The stages provide a shared temporal scaffold for cross-channel coordination. All three channels follow the same high-level affective progression and stage boundaries while retaining channel-specific content and intensity within each stage. Facial motion supplies embodied deformation beyond the neutral articulation baseline; regional facial color contributes spatially localized appearance cues; and symbolic effects provide discrete and explicit cues at semantically appropriate moments. This division allows the channels to assume complementary roles rather than requiring each channel to repeat the same affective signal. The planned facial-motion component is subsequently combined with neutral lip motion by the facial-motion fusion module, whereas regional facial color and symbolic effects remain separate control streams for rendering.

Because LLM outputs must be constrained before animation, a deterministic constraint layer validates the stage-level plan before realization. It enforces temporal consistency and valid control ranges, preserves the separation between articulation and emotional facial motion, and removes incompatible or conflicting channel outputs. The validated plan therefore retains the flexibility of semantic planning while providing bounded and independently controllable channel specifications for repeatable rendering and controlled comparison.

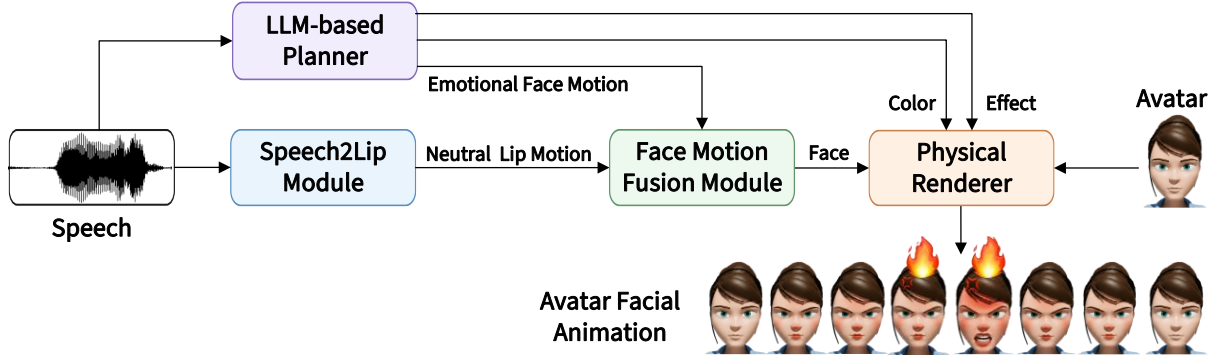


Figure 2: Overview of the speech-driven multi-channel avatar animation pipeline. The LLM-based planner generates stage-level plans for emotional facial motion, regional color modulation, and visual effects, while the Speech2Lip module produces frame-level neutral lip motion. The facial-motion fusion module combines the two motion sources into a complete facial-control sequence. The resulting facial motion, color controls, and visual-effect events are then executed by the physical renderer to generate the final avatar facial animation.

3.3 Prompt Design

Generating coordinated avatar behavior requires more than asking an LLM for an emotion label or a free-form performance description. We therefore design the prompt as a multi-round planning protocol that alternates between generation and self-reflection. In the first round, the model is assigned the role of an affective performance director for a stylized avatar and receives the spoken transcript, audio duration, a target number of stages, and, when required, an externally specified target emotion. Instead of predicting the three channels independently, it must first infer a shared affective progression and then instantiate facial motion, regional facial color, and symbolic effects from that progression. This coarse-to-fine organization keeps semantic intent above channel-specific realization and gives later reflection rounds a common structure against which to identify inconsistencies.

The planning prompt contains three groups of constraints. First, a temporal scaffold asks the model to organize the performance into pre-speech preparation, semantically ordered speaking stages, and an optional post-speech residual stage. Each speaking stage is anchored by a contiguous span copied from the transcript, and boundaries follow semantic pivots or affective turns rather than uniform time intervals. The model is also instructed to consider the preceding and following stages so that anticipation, reversal, peak, and carry-over form a coherent trajectory. Second, channel-specific instructions define a division of expressive responsibility. Facial motion is restricted to supported ARKit controls, while speech-critical jaw and mouth controls remain reserved for Speech2Lip. Regional facial color is represented by RGB values and activation weights over six facial regions, guided by stylized physiology-inspired patterns and emotion-dependent palettes. Symbolic effects are selected sparsely from a supported inventory with explicit placement, motion, timing, intensity, pairing, and mutual-exclusion rules. Third, a structured output contract requires synchronized stage arrays and a separate effect-event list, together with legal keys, numeric ranges, and a complete JSON example. The contract turns an open-ended semantic judgment into a plan that can be parsed directly by the animation pipeline.

After producing an initial plan, the LLM enters a self-reflection round in which the previous output and an explicit review checklist are provided as context. The checklist asks the model to inspect five aspects: temporal coverage and stage order; distinct development and placement of the expressive peak; semantic consistency across channels; complementarity rather than redundant cue repetition; and compliance with control ownership, supported assets, numeric ranges, and the required JSON structure. The model must first state the detected issues in a structured critique and then re-

vise the plan while preserving fields that already satisfy the constraints. The revised plan is fed into the next reflection round, allowing newly introduced conflicts to be detected rather than assuming that a single correction is sufficient. Iteration terminates when the model reports no remaining violations or when a preset round limit is reached.

This prompt design separates creative interpretation from explicit self-verification. The initial round retains the LLM’s ability to interpret nuanced utterances and construct an expressive trajectory, while the reflection rounds repeatedly test whether that interpretation remains temporally coherent, cross-channel consistent, and executable. Consequently, the final output is not merely an emotion description, but a self-checked stage-level specification that can be aligned with speech and rendered through the three visual channels.

3.4 Articulation-Preserving Facial Motion Fusion

Neutral lip motion and planned emotional facial motion serve complementary purposes but may overlap on expressive mouth controls. The former must preserve phonetic articulation, whereas the latter introduces affective deformation. Directly replacing the lip-motion sequence could weaken articulation, while unconstrained superposition could produce conflicting facial motion. We therefore introduce an articulation-preserving fusion module that integrates the two motion sources into the complete facial-control sequence $\mathbf{F}_{1:N}^{\text{full}}$.

The Speech2Lip branch uses NeuroSync [2] to produce frame-level ARKit-compatible facial coefficients [3], from which the articulation-related motion is retained as the neutral lip-motion sequence $\mathbf{L}_{1:N}$. In parallel, the stage-level emotional facial controls $\{\mathbf{f}_k\}$ are aligned with the speech timeline and converted into continuous frame-level targets $\phi_{1:N}$ using Smoothstep interpolation. A smoothed speech-energy descriptor r_t summarizes the local strength of articulation and guides the subsequent fusion without changing the shared stage structure.

We partition the facial controls into three disjoint ownership sets: articulation-owned controls Ω_{art} , shared expressive controls Ω_{shared} , and emotion-owned controls Ω_{emo} . The complete facial-control sequence is defined as

$$\psi_{t,c} = \begin{cases} \lambda_{t,c}, & c \in \Omega_{\text{art}}, \\ \beta(\lambda_{t,c}, \phi_{t,c}; \rho_t, \iota_t), & c \in \Omega_{\text{shared}}, \\ \phi_{t,c}, & c \in \Omega_{\text{emo}}, \end{cases} \quad (2)$$

where $\mathcal{B}$ denotes a bounded, articulation-aware blending operator conditioned on the speech-energy descriptor r_t and planned emotional intensity ι_t . The operator gives greater priority to neutral lip motion when precise articulation is required while allow-

ing emotional motion to contribute when it does not interfere with speech. Consequently, articulation-owned controls preserve the Speech2Lip output, shared controls balance articulation and affect, and emotion-owned controls follow the planned emotional motion. The resulting $\mathbf{F}_{1:N}^{\text{full}}$ forms the complete facial-motion channel rendered together with regional facial color and symbolic effects.

4 EVALUATION

We conducted two linked within-subject studies: Study 1 compares all Face–Color–Effect channel masks, and Study 2 compares four planning and timing configurations with all three channels active. Both used deterministic offline playback through the pipeline in Section 3; no LLM call occurred during a participant session. Joint Plan (JP) instantiates the proposed method, whereas the other Study 2 configurations are evaluation-specific ablations. Two non-overlapping cohorts of 36 participants were recruited from the same country. All reported familiarity with the Apple emoji conventions used by the Effect channel and normal color vision, and provided informed consent under prior ethics approval.

4.1 The Channel Composition Study

Prior work establishes Face, Color, and Effect as individually viable emotion channels, but separate evaluations do not show whether their cues remain complementary when presented together. Study 1 therefore uses a complete 2^3 factorial design to test every channel combination under matched speech, neutral lip motion, and avatar conditions. Applying the binary mask after validation also keeps the underlying joint plan fixed across masked versions of a base stimulus.

4.1.1 Participants and Design

The first cohort comprised 36 participants (18 women, 18 men; ages 18–25), all reporting English proficiency sufficient for the stimuli and questionnaire. Each completed 56 formal trials for 2,016 observations. Checks covered missing or out-of-range values, duplicate keys, correctness mismatches, and playback errors; no participant or trial was excluded. Eight counterbalancing lists contained four or five participants each. The $2 \times 2 \times 2 \times 7$ design crossed Face, Color, Effect, and target Emotion (Neutral, Happiness, Sadness, Anger, Fear, Disgust, and Surprise). Masks M000–M111 encode Face–Color–Effect. M000 retained the input speech waveform and neutral lip-motion sequence $\mathbf{L}_{1:N}$ but disabled all three planned visual emotion channels. In controlled mode, target peak intensity was fixed at 1.0 (Figure 3).

4.1.2 Stimuli, Counterbalancing, and Procedure

The 56 utterances were semantically neutral or weakly affective Harvard/IEEE sentences. The controlled target was supplied as separate #emotion= metadata and never entered the audible speech. Each base stimulus paired one utterance, target, character, waveform, neutral lip-motion sequence, and validated joint plan; the channel mask was applied afterward so masked versions shared the same plan. Each mask occurred 252 times, each emotion 288 times, each character 504 times, and each Mask×Emotion cell contained 36 observations. A participant saw one masked version of each base stimulus, and the eight lists rotated masks across stimuli.

Stimuli were shown on a Meta Quest 3 in a fixed VR scene with constant character placement, camera, lighting, background, and audio level. Playback ran on an AMD Ryzen 7 7735H workstation with an NVIDIA GeForce RTX 4060 Laptop GPU and 16 GB RAM. Each trial contained 0.5–1 s neutral lead-in, a 4–7 s performance, and an in-VR questionnaire, with one replay allowed. Four practice trials preceded four 14-trial blocks separated by 2–3 min breaks. Constrained randomization prevented the same emotion, mask, or character from appearing more than twice consecutively within a block.

4.1.3 Measures and Analysis

Recognition measured communicative accuracy and was coded 1 when the selected category matched the controlled target and 0 otherwise. Five-point confidence measured certainty in that judgment, and five-point perceived naturalness rated the performance from 1 (lowest) to 5 (highest). Naturalness was not a separate direct measure of cross-channel coherence. Replay and playback-error logs were diagnostic. Post-study overload, fatigue, and cue-use reports were analyzed descriptively because they were collected once per participant.

Recognition used a binomial mixed-effects model with Face, Color, Effect, their interactions, Emotion, and participant, base-stimulus, and character random intercepts; participant-clustered binary GEE checked robustness. Confidence and naturalness used participant-clustered ordinal GEE with linear mixed-effects sensitivity models. Repeated-measures ANOVA summarized factorial effects and Emotion interactions, with Greenhouse–Geisser correction where required. Planned participant-level contrasts used Holm-corrected paired t tests and Cohen’s d_z ; 95% intervals used 10,000 participant-cluster bootstrap samples (seed 20260823).

4.1.4 Recognition and Confidence Results

M111 produced the highest recognition (87.3%) and confidence (4.29/5), whereas M000 produced the lowest (44.0% and 2.75/5; Figure 4). The 43.3-point M111–M000 recognition difference was significant (95% CI = [34.1, 52.4], Holm-adjusted $p < .001$, $d_z = 1.52$). Face alone improved recognition over M000 by 28.6 points (95% CI = [20.2, 36.9], $p < .001$), and M111 exceeded M100 by 14.7 points (95% CI = [6.7, 21.8], $p = .002$, $d_z = 0.63$).

Face had the largest trial-level recognition effect (OR = 3.59, 95% CI = [3.06, 4.22]), followed by Effect (OR = 2.06, 95% CI = [1.78, 2.39]) and Color (OR = 1.46, 95% CI = [1.26, 1.70]); the Face×Effect OR = .72 (95% CI = [.57, .92]) indicates partial redundancy. Marginally, Face, Effect, and Color increased accuracy by 23.7, 10.8, and 8.0 points, respectively (all Holm-adjusted $p < .001$). Color improved the Face baseline (M110–M100: +9.1 points, $p = .019$), whereas the smaller Effect increment (+6.7, $p = .078$) did not reach significance for recognition, although it improved confidence. Confidence followed the same overall ordering, with marginal increases of 0.82, 0.47, and 0.25 points, respectively (all $p < .001$). Repeated-measures ANOVA corroborated the three recognition main effects and detected Face×Effect ($p = .035$), Color×Emotion ($p = .001$), and Effect×Emotion ($p = .007$).

Augmentation was emotion-dependent. Participant-clustered GEE also detected Face×Emotion, $\chi^2(6) = 14.59$, $p = .024$. Descriptively, Face improved every category, most strongly Surprise (+32.6 points) and Happiness (+30.6). Color contributed most to Anger (+23.6) but was slightly negative for Happiness (−5.6) and Neutral (−2.1); Effect contributed most to Sadness (+18.8) but was negative for Neutral (−9.0). M111 therefore did not improve Neutral over M000 (61.1% vs. 63.9%).

4.1.5 Naturalness and Post-Study Results

Naturalness had a different optimum: M100 scored 4.02 versus 3.51 for M111. M111 remained 0.22 points above M000 (95% CI = [0.10, 0.34], $p = .001$) but was 0.50 below M100 (95% CI = [−0.63, −0.38], $p < .001$). Marginally, Face increased naturalness by 0.55 points, while Color and Effect reduced it by 0.12 and 0.21 (all $p \leq .003$). Across the four tested characters, accuracy ranged from 64.5% to 71.4%, confidence from 3.61 to 3.67, and naturalness from 3.38 to 3.55. Character was modeled as a random effect and the principal channel estimates remained stable. Because character and base stimulus were not fully crossed and no Channel×Character test was planned, these summaries apply only to the tested allocations.

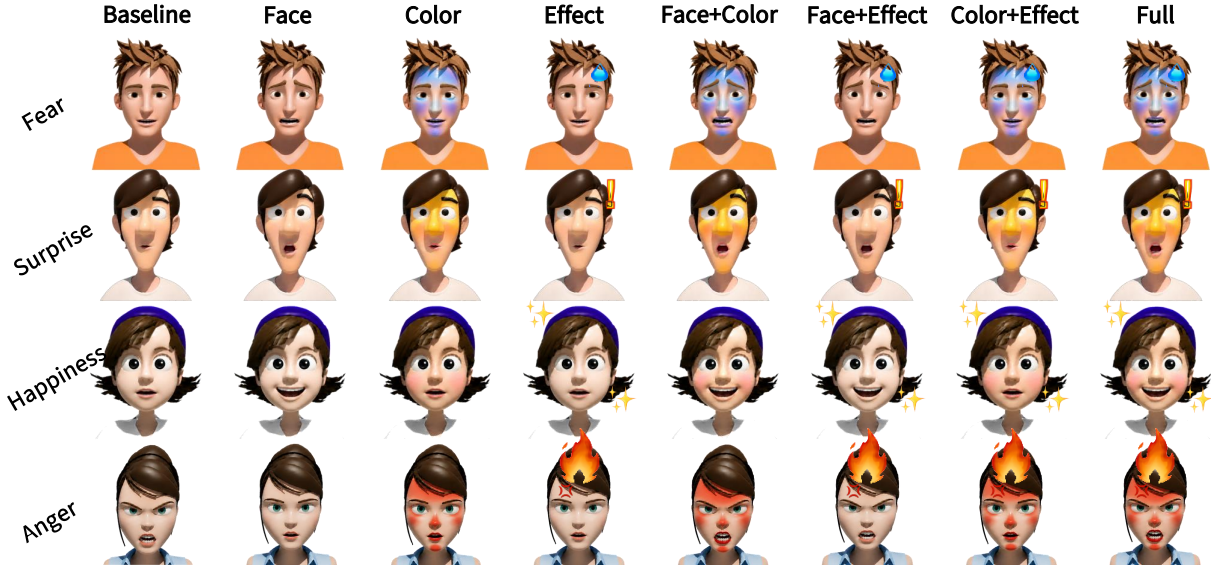


Figure 3: Representative Study 1 frames. Rows show Fear, Surprise, Happiness, and Anger; columns show M000, M100, M010, M001, M110, M101, M011, and M111. M000 retains the speech waveform and neutral lip-motion sequence while disabling the three planned visual emotion channels.

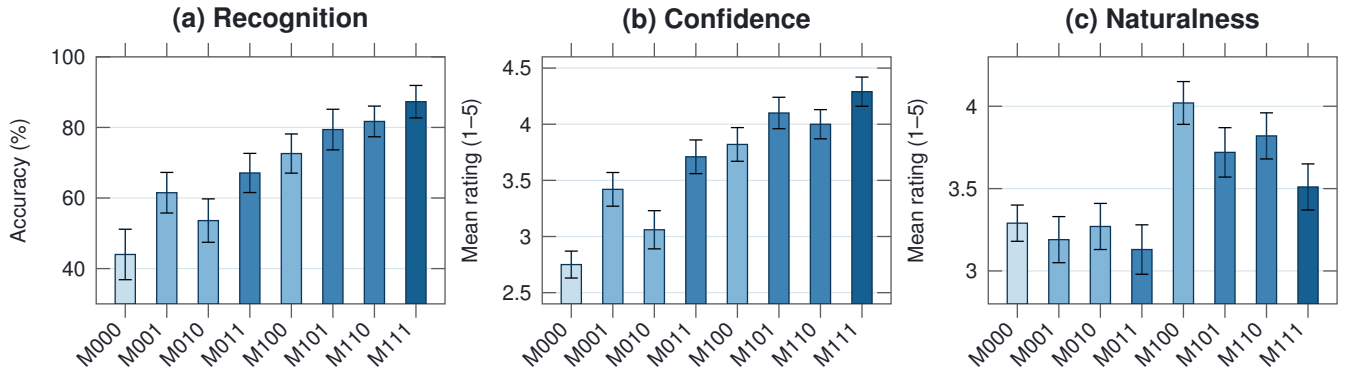


Figure 4: Study 1 outcomes across the eight Face-Color-Effect masks; error bars show 95% participant-cluster bootstrap CIs. Recognition and confidence peak at M111, whereas naturalness peaks at M100.

Post-study overload ($M = 2.81$, $SD = 0.82$, median 3) and fatigue ($M = 2.83$, $SD = 0.85$, median 3) were moderate-to-low. Face, Color, Effect, and Voice were reported as decision cues by 34/36, 22/36, 22/36, and 9/36 participants; 27/36 selected Face as most helpful. These participant-level reports contextualize the trial-level effects but do not estimate condition differences.

4.2 The Channel Coordination Study

Because M111 maximized recognition and confidence but remained less natural than M100, Study 2 keeps all channels active and compares four planning and timing configurations. The comparison tests whether configuration-level differences in joint versus separate planning, shared versus independently generated stages, and aligned versus shifted peaks are reflected in perceived naturalness while channel availability remains fixed.

4.2.1 Participants and Design

A separate cohort of 36 participants viewed 28 videos each (1,008 observations) in four nine-participant sequence groups that varied presentation order only. The 4×7 design crossed Condition with

the seven target emotions; every condition enabled Face, Color, and Effect. JP used the validated joint plan in Section 3. Shared Timeline (ST) retained JP stage boundaries but generated the three channel plans separately. Independent Timelines (IT) independently generated channel stages, content, and timing. Shifted Peaks (SP) copied its paired JP plan but displaced the three channel peaks.

SP preserved Face controls, Color parameters, Effect assets and events, and each channel's peak intensity, envelope shape, duration, and AUC; Effect event count was also fixed. Only temporal placement changed: peaks were assigned to 25%, 50%, and 75% of the target stage, rotating early/middle/late channel assignments across emotions. System logs recorded channel peak times, peak-time spread, intensity, AUC, and duration. JP-SP is a timing-specific comparison only when these quantities meet their prespecified tolerances.

4.2.2 Stimuli, Procedure, and Analysis

Seven base stimuli, one per emotion, yielded 28 fixed videos. Within an emotion, versions shared the target, text, speech waveform, neutral lip-motion sequence, character, scene, audio, dura-

Table 1: Study 2 outcomes. Accuracy and confidence remain descriptively close; ratings use five-point scales.

Condition	Accuracy	Confidence	Naturalness
Joint Plan	83.3%	3.85	4.12
Shared Timeline	83.7%	3.80	3.52
Independent Timelines	83.3%	3.83	2.94
Shifted Peaks	84.1%	3.81	3.12

Table 2: Prespecified response-scale naturalness contrasts from unrounded model estimates; all Holm-adjusted $p < .0001$.

Contrast	Targeted comparison	Δ [95% CI]
JP-ST	Joint planning	.60 [.53, .67]
ST-IT	Shared vs. independent plans	.57 [.49, .65]
JP-IT	Joint vs. independent plans	1.17 [1.10, 1.24]
JP-SP	Peak placement	1.00 [.93, 1.07]

tion, and validation rules; only configuration-defined planning or peak placement changed. Every participant saw each video once, giving 36 observations per Condition $\times$ Emotion cell, 252 per condition, and 144 per emotion. Four seven-trial blocks each contained every emotion once, with the four versions of an emotion placed in different blocks. For emotion index e , block b , and sequence group g , the assigned condition was $(e + b + g) \bmod 4$. Constrained randomization prevented the same condition from appearing more than twice consecutively. Four practice trials preceded the study, and 1–2 min breaks separated blocks.

Recognition used a logistic mixed-effects model with Condition, Emotion, their interaction, sequence group, trial position, participant intercepts, and Condition slopes. Confidence and naturalness used cumulative-link mixed models with the same structure. The four prespecified contrasts used Holm correction. Because each Condition $\times$ Emotion cell contains one fixed video, stimulus and script random effects cannot be estimated; accuracy and confidence are therefore described from their observed ranges, not treated as statistically equivalent.

4.2.3 Results

Recognition ranged from 83.3% to 84.1% and confidence from 3.80 to 3.85, whereas naturalness ranged from 2.94 for IT to 4.12 for JP (Table 1).

All four prespecified naturalness contrasts were significant after Holm correction ($p < .0001$; Table 2). JP exceeded ST by 0.60 points, ST exceeded IT by 0.57, and JP exceeded IT by 1.17. JP-ST compares joint with separate channel planning under shared boundaries; the difference may reflect channel content or relative emphasis produced by the separate plans. ST-IT compares a shared-boundary configuration with independently generated stages, content, intensity, and timing, so it is a configuration-level rather than timeline-only manipulation. The contrast estimates use unrounded model predictions, explaining small differences from subtraction of the displayed means.

JP exceeded SP by 1.00 naturalness point. Conditional on the logged tolerances, this comparison tests peak placement while retaining the paired JP Face controls, Color parameters, Effect content, intensity envelopes, and Effect event count. Because every configuration retained the same three channels and target category, recognition and confidence remained descriptively close while perceived naturalness differed substantially.

4.3 Evaluation Summary

Study 1 identifies M111 as the recognition/confidence optimum and M100 as the naturalness optimum. Face supplied the largest recognition gain and positive naturalness effect, while Color and

Effect provided smaller, emotion-dependent recognition benefits and reduced naturalness marginally. With all three channels active, Study 2 shows that planning and timing configurations can differ substantially in perceived naturalness even when their observed recognition and confidence remain close. JP was rated more natural than ST and IT, but these are configuration-level comparisons because separate or independent generation also changes channel content and emphasis. JP-SP is the narrower timing comparison, conditional on its logged tolerances. Together, the studies distinguish the observed effects of channel selection from those of the tested coordination configurations without treating naturalness as a direct coherence measure.

5 DISCUSSION

5.1 Complementary Channels Serve Different Roles

The channels are not interchangeable renderings of one label. Planned Face motion, layered on the retained neutral lip-motion backbone, produced the largest recognition gain and the only large positive marginal naturalness effect. This pattern is consistent with facial motion supplying the primary continuous expressive structure. Color and Effect added category-specific appearance and symbolic cues, but only Color significantly improved recognition beyond the Face-only condition; the smaller Effect increment did not, although it improved confidence. They are therefore better interpreted as potential communicative amplifiers than replacements for facial behavior.

M111 maximized recognition and confidence, whereas M100 maximized naturalness, revealing a multi-objective choice rather than one universally best mask. These outcomes need not agree: rapid or accessible emotion communication may favor more explicit cues, whereas sustained social interaction may favor a less stylized performance. The factorial masks establish candidate operating points under the tested stimuli, not a universal preference ordering. A future deployment policy could use the explicit channel plan to choose between them according to task and audience, but that policy was not evaluated here.

The partial Face $\times$ Effect redundancy and the naturalness decline from M100 to M111 also qualify the assumption that adding expressive cues is always beneficial. A cue can raise category evidence while duplicating information already conveyed by Face or increasing visual competition. Channel value should therefore be assessed incrementally against the current performance rather than by accumulating every available channel. This is especially important for symbolic effects, whose meaning may be clear while their visual style differs from anatomically grounded facial motion.

The effects also depended on emotion. Face improved every category, Color contributed most to Anger, Effect most to Sadness, and M111 did not improve Neutral. These descriptive patterns are compatible with the category-specific meanings of localized color and symbolic graphics discussed in Section 2. They support selective augmentation: Face can provide the continuous expressive structure, while Color or Effect may add information at a relevant stage. The six-region Color parameters and bounded Effect vocabulary make those contributions explicit in the execution plan. Because neither adaptive selection nor a general Character $\times$ Channel effect was tested, this remains design guidance rather than a validated online controller.

5.2 Coordination Differences in Perceived Naturalness

All four Study 2 configurations retained Face, Color, and Effect. Recognition and confidence consequently occupied narrow observed ranges, while naturalness varied from 2.94 to 4.12. This pattern is consistent with viewers responding to how the cues were organized after channel availability was fixed. However, naturalness was the measured construct; cross-channel coherence was not

rated separately, and the narrow accuracy and confidence ranges do not demonstrate statistical equivalence.

The contrasts must be interpreted at the configuration level. JP and ST shared stage boundaries but differed in joint versus separate generation, producing a 0.60-point naturalness difference. Their common boundaries control one part of temporal organization, but their channel content and relative emphasis can still differ because ST uses separate plans. The result therefore favors joint planning under the tested shared-stage setup without identifying which generated content differences mattered to viewers.

ST and IT differed by 0.57 points, but IT independently generated stages, content, intensity, and timing. The contrast consequently compares a shared-boundary, separately planned configuration with fully independent channel plans; it cannot isolate shared boundaries alone. The 1.17-point JP–IT difference likewise captures the combined configuration change. A future crossed ablation that holds channel-specific content and envelopes fixed while changing only stage boundaries would be needed to estimate a stage-structure effect independently.

JP–SP provides the narrower timing comparison. SP retained paired JP Face controls, Color parameters, Effect assets and events, and the channel intensity envelopes while moving their peaks; JP was rated 1.00 point more natural. Conditional on the logged intensity, AUC, duration, and Effect-event-count tolerances, aligned peaks were therefore more natural in these stimuli. The tested 25%–50%–75% displacement is only one offset pattern, so the result should not yet be treated as a universal synchrony law or an estimate of an acceptable timing window.

Together, these comparisons make stage organization and within-stage peak placement useful but distinct authoring variables. Shared stages describe the semantic progression of an utterance, whereas peak placement determines where each channel concentrates emphasis inside that progression. The current design gives stronger control for the latter through JP–SP than for the former through ST–IT. Both variables should remain explicit in the representation so future studies can manipulate them independently.

The JP result also illustrates the intended division of labor in the method. It supports generating complementary channel roles in one plan, not unconstrained animation generation. The planner produces the shared stage-level JSON, while deterministic validation clips values, repairs channel-list lengths, removes invalid facial controls or Effect events, assigns neutral values to missing Color regions, and aligns valid Effect events with their stages. The execution layer then constructs the frame-level streams and preserves neutral lip motion throughout. This combination keeps the semantic proposal inspectable and the rendered parameters bounded.

5.3 Design Implications for Social VR

The findings suggest a future two-layer policy for expressive social VR. The speech waveform and neutral lip-motion sequence should remain the articulation backbone, while planned Face motion supplies the main continuous visual expression. Regional Color and symbolic Effect can form an optional augmentation layer when their semantic role is clear. When Face already provides sufficient evidence, augmentation can remain subtle or inactive; when a category is ambiguous, one diagnostic cue could be placed at the relevant stage. The present studies motivate this policy but do not validate adaptive selection during live interaction.

The clarity–naturalness trade-off should also be treated as context-sensitive rather than reduced to one global optimum. Training, accessibility, or time-critical communication may reasonably prioritize explicit category cues, while sustained or intimate conversation may prefer a subtler appearance. M111 and M100 offer empirical candidate settings for these ends in the present stimulus set. Determining when to switch between them would require task-level outcomes and user preferences that were not collected here.

The exported shared-stage JSON and repair record could support a future authoring interface. A designer could inspect a short sequence of stage boundaries and channel-specific controls, adjust regional Color parameters or discrete Effect events, and retain the deterministic validation constraints. Because Face controls, Color regions, and Effect events remain typed separately, edits need not be made as independent frame tracks. This interface and its usability were not evaluated; the current implementation validates, exports, and replays generated plans offline.

Character-aware calibration is another extension. Outcomes differed descriptively across the four avatar–stimulus allocations, but character and base stimulus were not fully crossed, so the study cannot establish a Character×Channel interaction. A fully crossed design could test whether per-character control ranges, Color-region visibility, Effect attachment offsets, or intensity ceilings improve performance while retaining one semantic plan. The same logic applies to display conditions and prior VR experience: the current results identify design options, not the socially appropriate choice for every avatar or setting.

6 CONCLUSION

We presented a speech-driven, LLM-assisted, rule-constrained apparatus for combining and coordinating facial motion, regional facial color, and symbolic effects while holding speech and neutral lip motion constant across conditions. The Channel Composition Study compared all eight channel combinations and found that the three-channel condition achieved the highest emotion recognition accuracy, whereas the Face-only condition received the highest naturalness ratings, revealing a trade-off between affective legibility and perceived naturalness. The Channel Coordination Study kept all three channels active and compared four planning and timing conditions. Cross-channel coordination improved perceived naturalness while leaving recognition accuracy and judgment confidence largely unchanged. Together, these findings distinguish the perceptual roles of channel composition and coordination and provide controlled evidence for combining and coordinating visual channels in speech-driven virtual humans.

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